Module 3 — The Query Library

45-minute block · TIP4PATLIBS course material

> **How to read this document.** The running text addresses **you, the participant**. Boxes labelled **Trainer** are for whoever is **running the session**; boxes labelled **Trap** are traps that have caught people before.

Learning objective

I can pick a ready-made PATSTAT query, adapt its parameters to my own question, and judge whether the result is plausible — without writing SQL.

Prerequisites

- **Module 1** — TIP is running, PATSTAT connects with `PatstatClient(env='PROD')`.
- No SQL knowledge. That is the point of this module.

Sub-objectives

By the end you can:

1. **Map a client's question onto a query category** — decide *which* of the library's queries answers what you were actually asked.
2. **Change parameters safely** — country, year window, technology field — and know which changes are cheap and which are expensive.
3. **Judge plausibility** — recognise a result that is suspiciously large, suspiciously small, or silently truncated.
4. **Know when to stop** — recognise the questions the library cannot answer, and what to reach for instead.

Material

Folder	`3_querylib/`
Notebooks	`TIP_for_PATLIBs_QueryLib.ipynb` (the browser) · `TIP_for_PATLIBs_InteractiveQueryDemo.ipynb` (point-and-click)
Runs on	EPO TIP, PATSTAT PROD

Phase 1 · Introduction (≈ 7 min)

The question this module answers

> *A client walks in: "Who in Europe is working on solid-state batteries?"*

Give the room two minutes to say how they would answer it. Collect the proposals.

What almost always comes back first is a keyword search. And a keyword search is not wrong — it is just not defensible. Ask the follow-up: *how would you prove to that client that your list is complete?* Keywords miss everything phrased differently, and catch everything that merely mentions the term. Neither problem is visible in the result.

That gap is what a curated query library exists to close. Someone has already decided which tables to join, which filters are non-negotiable, and what "complete" means for each type of question. Your job shifts from *writing* the query to *choosing* it and *reading* it.

	Teaching and learning activity	Time
Opening	Trainer poses the client question; room proposes approaches out loud	2 min
Tension	Trainer asks "how do you prove it is complete?" — the proposals fall apart	2 min
Framing	Trainer introduces the library as *pre-made professional decisions*, not as a shortcut	3 min

> **Trainer.** The expected wrong answer is *"search Google Patents"* or *"type keywords into Espacenet"*. Take it seriously — it is what most people do and it is often adequate. Do not mock it; ask what happens when the client's competitor turns up a patent that your list missed. The room will get there on its own.
>
> **Trainer.** If the room is quiet, name a concrete failure: a search for *"battery"* misses every document that says *"accumulator"* or *"electrochemical cell"*.

Phase 2 · Working through (≈ 28 min)

Step 1 — Open the library and look at what is there (8 min)

Run `TIP_for_PATLIBs_QueryLib.ipynb` top to bottom. It has only two cells: a setup cell that initialises the registry, and the **Query Browser**.

In the browser:

- **Browse by category** using the dropdown.
- **Search** by typing a keyword — it searches titles, descriptions and tags.
- **Preview** a query to see what it needs and what it returns.

- **View SQL** to see the template with its parameters highlighted.

Do this deliberately: open "View SQL" at least once. You are not expected to read it fluently. You are expected to see that there *is* something to read — that the number the library gives you comes from a statement you could inspect, hand to a colleague, or argue about.

> **Trainer.** This is the moment worth slowing down for. The difference between a tool and a black box is whether you can open it. Say it out loud: *an analysis you cannot open up is one you cannot defend.* The same sentence returns in modules 4 and 8.

Step 2 — Adapt one query to your own question (12 min)

Pick a query that matches a question you actually get asked. Change its parameters — country, year window, technology field — and run it.

Two rules while you do this:

- **Change one thing at a time.** If the result surprises you, you want to know which change caused it.
- **Watch the clock on the query.** The library reports how long each query took.

> **Trap — PATSTAT PROD is not fast.** A query scanning a full year of filings takes tens of seconds to a minute or more. That is normal, not a fault. Widening a year window or dropping a country filter multiplies the work; a query that ran in 10 seconds can take several minutes with the filter removed. Say this to the room *before* someone decides the notebook has crashed.

Step 3 — The interactive demo (8 min)

Open `TIP_for_PATLIBs_InteractiveQueryDemo.ipynb` and run the first cell to load the components. Then use the **Selection** controls — jurisdiction, region, technology field or custom IPC/CPC, date range — and read the **Results**: a table and a chart, no code touched.

This is the same data through a different door. Note which door you would put in front of a client, and which you would use yourself.

> **Trap — Stop after the "Data Validation" section.** The notebook ends with two cells headed *"Story 3.2 AC Validation"* and *"Story 3.3 AC Validation"*. Those are developer acceptance tests left over from building the tool — not course content. Skip them; they will run, and their output means nothing to a participant.
>
> **Trainer.** Note this as a known rough edge rather than apologising for it. It is honest, and it is on the cleanup list.

Step	What you do	What you see	Time
1	Browse the library, open one query's SQL	The decisions someone made for you	8 min
2	Adapt parameters, run, time it	Your own question answered — and what it costs	12 min
3	Drive the interactive demo	The same data through a point-and-click door	8 min

Phase 3 · Learning outcome (≈ 10 min)

What now exists

- One query adapted to a question **you** are actually asked, with its result table.
- A first feel for what a PATSTAT query costs in time.

Self-check

Answer these before moving on. If one is unclear, go back to the query that raised it.

1. **Which query would you use** for *"has patenting in my region gone up or down over ten years?"* — and which would you use for *"who are the biggest filers in this technology?"*
2. **Why does the same question return fewer rows** when you filter `appln_auth = 'EP'` than when you count `docdb_family_id`? *(Because one counts filings at one office, the other counts inventions wherever they were filed. Module 4 makes this the centre of the lesson.)*
3. **You widened the year window from 5 to 20 years and the query is still running after three minutes.** Broken, or working? *(Working. Narrow it, get the answer, then widen deliberately.)*
4. **A client asks which of two companies has the "better" portfolio.** Can the library answer that? *(No. It can measure size, reach and technology mix. "Better" is a judgement — and module 8 is about exactly that boundary.)*

Transfer to your own work

Write down the **three questions you are asked most often** at your PATLIB desk. For each, note which library query comes closest, and what is missing. That list is your agenda for modules 4 and 5 — and it is worth keeping.

> **Trainer.** Close by collecting two or three of those questions from the room. If one of them is *"who around here is patenting?"*, you have your bridge into module 5. If it is *"is this company big or small in IP?"*, that is module 4.

Where this leads

Next	Why
Module 4 — PATSTAT Explorer	The library counts *names*. Module 4 shows why one company is forty different names, and how to consolidate them.
Module 5 — Lead Generation	The first full application: a region's applicants, ranked and segmented.

Notes for the next revision

- The two *"Story 3.x AC Validation"* cells at the end of the interactive demo are developer scaffolding and should be removed from the course version of the notebook.
- Module 3 is the shortest module in the course (≈ 8 minutes of notebook content). This block works because the introduction and the transfer exercise carry real weight — do not cut them to save time.